🏢 Staffing Agency RL Environment

Agentic RL — System Design Flow

🏦 CLIENT

- client_id, industry
- contracted_rate: $100K/yr
- satisfaction_score
- churn_risk
- projects[] (multi-role)

submit_project(rol →

🏢 STAFFING AGENCY (Agent)

- cash_balance
- num_hired / placed / benched
- placement_rate
- avg_time_to_place
- current_profit

find_candidate(typ →

👤 CANDIDATE POOL

- developer_type
- skill_score
- seniority_level
- salary: ~$80K/yr
- patience_remaining
- status: avail/hired/placed

🔍 GET TOOLS (State)

- get_agency_state()
- get_client_state()
- get_candidate_state()
- get_project_details(id)
- get_market_demand()
- get_financial_summary()

observe state - - - →

🧠 AGENT DECISION

- Policy π(obs) → action
- Observation vector:
  - cash, profit, bench
  - projects (per client)
  - role demand breakdown
  - churn risk flags

execute action →

⚡ EXECUTE TOOLS

- find_available_projects()
- confirm_project(id)
- find_candidate(type)
- interview_candidate(id)
- hire_candidate(id)
- match_candidate_to_project(
- let_go_candidate(id)
- advance_week()

📋 PROJECT (Multi-role)

- project_id, client_id
- roles[]: {role_id, dev_type,
  - seniority, headcount}
- deadline_remaining
- fill_status: PARTIAL/SEALED
- bill_rate_weekly (per role)

seal when all roles filled →

⚖️ REWARD ENGINE

- +margin/wk per placed
- −salary/wk per benched
- −$500 (interview)
- −$2K onboarding (hire)
- −2×salary severance
- −expiry + churn penalties
- +speed seal bonus

update policy →

📈 RL TRAINING LOOP

- obs → policy → action
- env.step(action)
- → next_obs, reward, done
- PPO / GRPO update
- Episode: 52 steps (1yr)
- Terminal: bankrupt or T_max

💳 **Payment flow:** submit_project → confirm → match all roles → SEALED → bill_rate_wk flows to agency → agency pays salary_wk → **margin = bill_rate − salary**

A single project has multiple roles. Revenue only unlocks when ALL roles are filled.

📋 Project (multi-role)

project_id: P001 | Client: Acme Corp

Roles Required:

- role_1: Backend Dev × 2 ✓ 1/2
- role_2: ML Engineer × 1 ✗ 0/1
- role_3: DevOps × 1 ✓ 1/1

deadline_remaining: **3 weeks**
fill_status: **PARTIAL (2/4)**
Revenue trigger: when 4/4 filled

↓

Agency must fill each role_id separately

role_1
type: **Backend Dev**
headcount: **2**
min_skill: **0.70**
seniority: **Mid/Senior**

role_2
type: **ML Engineer**
headcount: **1**
min_skill: **0.80**
seniority: **Senior**

role_3
type: **DevOps**
headcount: **1**
min_skill: **0.65**
seniority: **Mid**

↓

All roles filled → SEALED → bill_rate_wk revenue begins

Partial fills (2 of 4) = **$0 revenue**. Agent bears full salary burn on all benched candidates while filling.

How the agent maps a candidate to a specific role inside a project.

🔗 Matching Logic

match_candidate_to_project()

Match Score Criteria:

- ✅ developer_type == role.type
- ✅ skill_score ≥ role.min_skill
- ✅ seniority compatible (senior ≥ mid ≥ junior)
- ⚠️ adjacent type → 0.7 score penalty

params: candidate_id, project_id, role_id

💰 Revenue only when ALL roles filled

✅ Perfect Match

```
type == role.type
skill ≥ min_skill
seniority OK
```

match_score: 1.0

⚠️ Adjacent Match

```
e.g. Fullstack →
Frontend
Partial skill overlap
Seniority overqualified
```

match_score: 0.7–0.85

❌ No Match

```
type mismatch (non-
adjacent)
skill_score < min_skill
junior filling senior
role
```

match_score: 0.0 (BLOCKED)

Adjacent matches reduce client **satisfaction_score**, increasing churn risk even if the project seals. Optimal policy prefers perfect matches.

Reward decomposition — actual implementation from environment code.

⚖️ Reward at advance_week()

+ Σ (bill_rate_wk − salary_wk) per placed candidate

− Σ salary_wk per benched candidate

− bill_rate_wk × weeks_remaining (per expired role)

− estimated LTV (per churned client)

📐 Margin varies: ~$200–$1,500+/wk per head

📐 Bench cost: full salary_wk (continuous, not fixed)

Placed Candidate

margin: **+varies**

bill_rate_wk − salary_wk. Continuous — depends on role scarcity + client premium.

Benched Candidate

weekly: **−salary_wk**

Full salary burn. No revenue. Most dangerous state for the agent.

Hire Event

cost: **−$2,000**

One-time onboarding. Immediate deduction on hire_candidate().

Interview Event

cost: **−$500**

Per interview_candidate() call. Screens skill + salary via LLM.

Let Go Event

cost: **−2× salary_wk**

Severance. Higher-rated candidates cost more to release.

Project Expired

penalty: **−bill_rate × wks**

Missed billing opportunity. Also reduces client satisfaction.

⚡ **Behavioural penalties:** invalid action, repeat call, passive streak — prevent degenerate policies.

🏆 **Speed seal bonus:** +10% margin if project sealed within 2 weeks of confirmation.